Semantic Chemistry: Algorithmic Chemistry on Semantic Graphs as a Creative Inference Control Strategy

Ben Goertzel and GPT-5.5 Pro

May 7, 2026

Abstract

This document sketches a research program that brings together three threads that have rarely been combined: algorithmic chemistry as a substrate for adaptive computation, semantic graph processing as a way of representing meaning, and Probabilistic Logic Networks (PLN) as a calculus of uncertain inference. The basic proposal is to build an atomspace of variable-bearing semantic graphs and to construct a weighted rule soup of graph transformations of the form `G1 ==> G2`, or more generally `C ^ G1 ==> G2`, where `C` is a context graph. The weight of such a rule is estimated using PLN intensional implication: roughly, the degree to which the properties, patterns, roles, or consequences associated with `G1` are also associated with `G2`, possibly relative to `C`. These rules are not used merely as a static inference base, in the way classical logic programs are used. They are placed in an algorithmic chemistry in which rules, graph fragments, contexts, answer skeletons, and learned macro-patterns interact as molecules, catalysts, food, and reaction products. The metaphor is meant literally: just as chemistry studies which molecular species persist, react, or vanish in a given environment, this framework studies which patterns of inference persist, react, or vanish in a given semantic context.

For natural language applications, sentences are parsed and normalized into semantic graphs, for example via a FUSE-NF-like pipeline that combines syntactic structure, predicate-logic structure, statistical consolidation, background PLN inference, causal labels, and graph-transformer encodings. The resulting semantic graphs are treated as molecules. Contextual rewrite rules are reactions. Context chambers localize the reaction ecology, much as compartments localize biochemistry in a cell. ECAN budgets attention and runtime, playing the role of an energy economy. ACS/RAF detection identifies self-sustaining semantic inference loops; these are the analogs of autocatalytic cycles in real chemistry. External reward, such as successful question answering or interesting fiction generation, selects useful creative chemistries from among the many that the system might otherwise sustain.

This document is intended as a stand-alone follow-on to the longer ActPC-Chem / MORK architecture document. It assumes that document's basic execution picture: MM2 chambers provide ordered local reaction traces; MORK provides indexed atom storage; ECAN provides attention allocation; reducers and worker orchestration provide scalable execution; and useful active autocatalytic sets are the central scientific objects. Here the focus is not the backend mechanics, but the semantic-graph chemistry that can run on top of that substrate.

A next step is to analyze semantic graph chemistry in terms of causal coding. PLN intensional implication supplies plausible semantic transformations. The chemistry explores loops of these transformations. Causal coding then estimates which rules, modules, chambers, and ACSs have interventional influence on downstream reward, prediction error, answer quality, biomedical hypothesis utility, or narrative coherence. This adds a modularity and gating discipline to the loopy rule soup: exact global double-counting control is not required, but locally redundant or non-interventional loops must pay rent, lose clarity, or fail ablation. The intuition is that a

thriving inference ecology may contain many redundant pathways without harm, but it should be honest about which pathways actually carry causal weight.

Contents

1 Motivation and relation to companion documents

This is an addendum to a longer companion document, ActPC-Chem on MORK, which proposes an algorithmic-chemistry substrate in which weighted rewrite rules, traces, attention, structural evolution, and autocatalytic-set detection interact. The claim central claim of this longer document is not merely that a system can learn rules in the usual sense. It is that a system can discover *useful active autocatalytic sets*: recurrent networks of rules, intermediate artifacts, program fragments, or other structures whose self-supporting organization is causally responsible for useful external behavior [1]. This shifts the scientific question from "what is the right rule?" to "what recurrent network of rules and intermediates keeps reproducing itself because it works?".

The present document takes that architecture into a new domain: semantic graph inference. In this setting the molecules are not raw program fragments or sub-symbolic features but symbolic representations of meaning. The proposed semantic chemistry can be used for natural language, where semantic graphs come from sentence parsing, discourse parsing, and semantic normalization. It can also be used in nonlinguistic domains wherever one has enough data to estimate intensional implication links between variable-bearing subgraphs. The same engine that learns "rain causes driving" from text could in principle learn "this gene expression motif tends to imply this phenotype" from biomedical pathway data, or "this proof-state pattern tends to imply this lemma form" from mathematical corpora.

Five prior ideas are used as ingredients. Each addresses a different aspect of the problem of building creative, structured inference under uncertainty.

1. The ActPC-Chem / MORK document supplies the algorithmic-chemistry substrate: local reaction chambers, hot pools, event logs, ECAN attention, worker-level orchestration, structural rule evolution, and ACS/RAF detection [1]. This is the "machinery" on which the semantic chemistry runs.

2. The PLN framework supplies weighted uncertain inference, especially truth values, intensional implication, contextual inference, uncertain abduction, analogy, speculation, causal reasoning, and the observation that inference control is a major unsolved practical issue in uncertain reasoning [2]. PLN tells us how to attach meaningful weights to graph rewrite rules and how to combine them when they overlap.

3. The FUSE-NF concept supplies a semantic-graph normal form for natural language: a pipeline that combines CCG-like syntactic semantics, LLM-generated predicate-logic interpretations, pattern-mined consolidation, background PLN inference, causal or correlational edge labels, and graph transformer representations [3]. This provides the stable molecular shapes the chemistry will react over. Without something like FUSE-NF, the chemistry would constantly waste energy treating paraphrases of the same idea as unrelated objects.

4. The causal-coding framework supplies a modularity principle for continual learning. Contexts induce update operators; if the learner factors state into causally meaningful modules and updates mainly the modules that matter in the current context, then different context updates have small commutators. Sequential learning is then close to joint learning, with error controlled by genuinely confusing context pairs rather than by all pairs of contexts [4]. In other words, causal coding offers a principled story about why learning one thing need not destroy what was learned earlier, provided the system's internal modules line up with the causal structure of the world.

5. The independence-coupling tradeoff supplies a complementary resource view. A system can either spend effort on Route A, independence-enforcing transforms such as normalization, modularization, and gating, or Route B, explicit dependency modeling such as implication links, causal edges, attention links, and ACS couplings. Practical intelligence lives on a Pareto frontier between these two routes [5]. Cleaning up dependencies is sometimes cheaper than modeling them, and modeling dependencies is sometimes cheaper than cleaning them up; a real system needs both, in the right proportions.

These five ingredients fit together into a single pipeline. At a glance:

```
FUSE-NF / semantic parsing:
    create normalized semantic graph molecules

PLN intensional implication:
    estimate weighted semantic graph rewrite rules

Algorithmic chemistry:
    evolve, select, combine, and replay semantic inference-control loops

Causal coding:
    estimate interventional influence, gate modules, and damp redundant loops

External tasks:
    reward useful question answering, discovery, explanation, or fiction
```

The high-level promise is that creativity can emerge from structured, repeated, context-sensitive graph transformations, rather than from either linear proof search or surface-level textual imitation. Linear proof search is too brittle and too narrow to be creative; surface-level imitation lacks the structural understanding needed to invent, explain, and revise. Semantic chemistry occupies a middle ground in which inference is both structured and exploratory.

2 General concept: PLN-weighted graph rewrite chemistry

2.1 Graph patterns as terms

Before describing rules, we need to be clear about what they are rules over. Let `G` denote a variable-bearing graph pattern. In the natural language case, `G` may be a semantic graph fragment containing event nodes, entity nodes, role edges, modality nodes, temporal edges, causal edges, and contextual annotations. In another domain, `G` may be a molecular interaction graph, a code graph, a robotic scene graph, a social graph motif, or a theorem-proving proof-state fragment. The point is that the same machinery applies whenever the domain can be encoded as labelled graphs with variables.

The graph pattern is treated as a term-like object, in the sense familiar from term-rewriting and logic programming. But unlike a flat term, a graph pattern carries two simultaneously useful aspects:

- an extension: the set of matches or instances of the pattern in a dataset or atomspace;
- an intension: the set of properties, associated patterns, roles, consequences, contexts, causal tendencies, or task utilities associated with that pattern.

Extension answers "what concrete things are instances of this pattern?". Intension answers "what does it tend to mean, predict, enable, or imply when this pattern shows up?". This distinction is directly aligned with the PLN distinction between extensional and intensional relationships. Extensional relationships compare terms according to members or instances. Intensional relationships compare terms according to properties or patterns associated with the terms [2]. The chemistry to follow will lean heavily on intension, because intension is what allows weak signals to combine into useful generalizations.

2.2 Semantic reaction rules

The basic rule form is:

```
G1 ==> G2
```

read as "a graph matching pattern `G1` can react to produce, or support adding, a graph matching pattern `G2`". More generally we allow a context to modulate when the rule fires:

```
C ^ G1 ==> G2
```

where:

```
G1:
    variable-bearing semantic graph pattern on the left hand side

G2:
    variable-bearing semantic graph pattern to add, strengthen, revise,
    project, compress, explain, or instantiate

C:
    context graph, such as a domain frame, question frame, discourse frame,
    source frame, causal frame, fictional-world frame, or reasoning style

weight:
    PLN truth value for IntensionalImplication(G1, G2), possibly within C
```

The context term C is essential. Many semantic implications are true only in certain frames: "X drove because of weather" is a strong implication in a why-question chamber, weak in a pure travel-description chamber, and irrelevant in a geographical chamber. Without contexts, the chemistry would have to choose between brittle global rules and an unmanageable proliferation of unconditional ones.

A schematic MORK-like representation makes the rule's identity, content, context, and weight explicit:

```
(sem-rule R44)
(rule-lhs R44 G_lhs_44)
(rule-rhs R44 G_rhs_44)
(rule-context R44 C_weather_why)
(rule-tv R44 intensional-implication 0.78 0.61)
```

Here `0.78` is a strength-like estimate (how reliable the rule is when it does apply) and `0.61` is a confidence-like estimate (how much evidence backs that strength estimate). A deployed system might instead use PLN indefinite truth values or distributional truth values, which carry richer information. The simple pair is shown only to make the architecture easy to read.

2.3 Estimating rule weights from a dataset

Where do these weights come from? In principle, from any source of evidence about how patterns relate to one another. In practice, the simplest approach is to mine them from a corpus of normalized graphs.

Let D be a dataset of semantic graphs. In natural language, D may be a corpus of parsed and normalized sentences. In biomedical science, D may be a corpus of abstracts, mechanistic graphs, clinical-trial summaries, pathway diagrams, and phenotypic annotations. In fiction, D may be a corpus of world rules, plot events, character arcs, myths, conflicts, and story outcomes. Each of these gives a different flavor of intensional implication; the same formal mechanism applies in each.

For a graph pattern G, define Pat_C(G) as the weighted set of properties or patterns associated with matches of G in context C. Then one simple intensional implication strength can be written:

$$s_C(G_1, G_2) = \frac{\sum_{p \in Pat_C(G_1) \cap Pat_C(G_2)} w_C(p)}{\sum_{p \in Pat_C(G_1)} w_C(p) + \epsilon}.$$

The numerator counts (with weights) the properties shared between the left-hand side and right-hand side patterns. The denominator normalizes by the total weight of properties associated with the left-hand side, so that the result is the fraction of the left-hand side's intensional "footprint" that the right-hand side covers. This is not proposed as the only formula. It is a compact way to express the concept: G1 intensionally implies G2 to the extent that the properties associated with G1 are also properties associated with G2. The confidence term can depend on support counts, diversity of sources, cross-context stability, provenance, and evidence quality. A simple count-based confidence is:

$$c = \frac{N}{N + k},$$

where N is an effective support count and k is a lookahead or smoothing parameter. This is the familiar shape that approaches one as evidence accumulates. PLN's richer indefinite truth values can be used when one wants intervals, credibility levels, and second-order uncertainty rather than a single confidence value [2].

2.4 From inference rules to a rule soup

A conventional inference system might store these implications and use them in forward or backward chaining: pick a goal, find rules that conclude it, recursively prove their premises. This produces clean proof trees but tends to be brittle, because real reasoning is full of weak overlapping evidence and benefits from many partial bridges that no single proof tree captures cleanly.

The proposed chemistry instead places the rules in an adaptive reaction ecology, where many rules can be active at once and where the population dynamics of the rules and intermediates becomes the central computation:

```
molecules:
    semantic graphs
    contexts
    question graphs
    answer skeletons
    learned graph rewrite rules
    PLN implication links
    paraphrase meta-nodes
    causal motifs
    ACS macro-atoms

reactions:
    graph rewrite
    graph expansion
    graph compression
    causal alignment
    role completion
    analogy mapping
    paraphrase consolidation
    answer projection
    explanation packing

catalysts:
    contexts
    high-STI semantic motifs
    successful answer templates
    discourse frames
    causal schemas
    ACS macro-atoms

food:
    parsed sentences
    background KB facts
    primitive semantic roles
    initial FUSE-NF fragments
    question graphs

reward:
    correct answers
    useful explanations
    novel hypotheses
    interesting story events
    compact justifications
    human or automated evaluator scores
```

Notice that the same kind of object can play different roles depending on the situation. A context graph can be a molecule when reasoned about and a catalyst when used to gate other rules. A successful answer template becomes a catalyst the next time a similar question arrives. This fluidity of role is part of what makes the chemistry analogy useful: in real chemistry too, the same molecule can be substrate in one reaction and catalyst in another.

The system does not merely ask which single rule is valid. It asks which semantic reaction networks become active, self-sustaining, useful, novel, and transferable. This is a population-level question, more like ecology than like deductive logic.

2.5 Inference control as the object of evolution

The shift from rules to ecologies has a deeper consequence. PLN describes uncertain inference rules and truth-value formulas, but practical uncertain inference requires inference control: which inferences should be made, in which order, in which context, and with how much resource budget [2]. In most uncertain reasoning systems, inference control is the bottleneck: the rules themselves are fine, but knowing when to apply which rule is the hard part. Semantic graph algorithmic chemistry treats this not as a side problem but as the central adaptive process.

Instead of:

```
run PLN with a fixed inference-control policy
construct a clean proof tree
try to avoid all double-counting exactly
```

we have:

```
run semantic graph reaction chambers
allow many approximate, repeated, overlapping transformations
log enough provenance for damping and replay
let ECAN, ACS detection, reward, and ablation select useful loops
```

This is like a loopy Bayes net, but more radical. In a loopy Bayes net the loopiness is a tolerated approximation: we would prefer no loops, but we accept them and try to bound the damage. Here the loopiness is not an accidental defect of an otherwise clean inference system. It is the substrate on which creative inference-control strategies are discovered. Recurrent inference patterns that prove useful in many contexts will become salient and reusable; the system's "cleverness" is largely the identification of such patterns, not the avoidance of loops.

2.6 Evidence double-counting

A common worry about the loopy rule-soup view is that it will produce inflated confidences by double-counting the same evidence along multiple paths. This is a real concern, but the usual response (insist on exact de-duplication) is unworkable in a creative inference ecology.

The proposal therefore deliberately avoids making exact global double-counting control a hard requirement. This is a design choice, not an oversight. In a creative semantic chemistry, redundant pathways can be useful. They can help motifs become active, allow weak signals to combine, and generate surprising bridges. Two paths to the same conclusion are sometimes the very thing that lets a weak hypothesis cross a threshold of attention.

However, the system should not be blind to circularity. It should log and damp it, using a layered set of pressures rather than a single exactness constraint:

```
attention rent:
    loops consume STI and CPU budget if they keep firing
```

```
truth-value confidence decay:
    low-diversity or heavily circular support has lower confidence

provenance trails:
    event logs remember which rules and graph matches contributed

duplicate-rule merging:
    many near-identical rules are collapsed or taxed

ACS ablation penalties:
    a loop must prove that removing it hurts task performance

contextual hot-pool limits:
    only a bounded set of rules is active in a chamber
```

Each of these is a soft pressure rather than a hard prohibition. Together they push the system toward parsimony without forbidding the exploratory redundancy that creativity often needs. The design principle can be stated compactly:

```
Do not try to eliminate all evidence double-counting.
Do log enough provenance that bad loops can be taxed, damped, or ablated.
```

3 Causal coding for semantic graph chemistry

3.1 Why causal coding is needed

The previous section deliberately allowed loopy, redundant, repeated inference for the sake of creativity. This raises an obvious follow-on question: how do we tell the useful loops from the merely active ones? Without an answer, the system risks reinforcing patterns that look internally coherent but do not actually help with anything.

Semantic graph chemistry deliberately permits loopy, redundant, repeated inference. This is useful for creativity, but it raises an obvious danger: a semantic rule soup may reinforce correlations that do not matter interventionally. Causal coding adds a modularity and gating layer that asks which rules, modules, workers, and ACSs actually influence downstream reward, prediction error, answer quality, hypothesis utility, or narrative coherence. The test it imposes is operational rather than statistical: does the system actually do better when this loop fires than when it does not?

The basic division of labor between the three layers is:

```
PLN intensional implication:
    tells us which semantic graph transformations are plausible.

ActPC-Chem:
    explores and evolves loops of such transformations.

Causal coding:
    asks which transformations have interventional influence,
    then gates attention, update, replay, and structural evolution accordingly.
```

The point is not to replace PLN truth values. PLN intensional implication is about shared properties, patterns, and variable-bearing semantic regularities. Causal coding is about whether using, perturbing, or ablating a rule changes what the system can do. These can come apart in both directions: a rule can be semantically plausible but causally irrelevant to a task, and a weaker rule can be causally important if it opens a bridge that changes the downstream state of the chemistry. The two assessments answer different questions, and a healthy system needs both.

3.2 Mapping semantic chemistry to causal coding

Causal coding was originally formulated for general continual-learning systems, but the mapping to the present setting is straightforward:

Semantic chemistry object	Causal-coding interpretation
Semantic graph	State component or molecule.
PLN intensional implication	Plausible context-conditioned reaction rule.
Context `C`	Task, domain, discourse, world, or question condition.
Reaction chamber	Context-local update operator.
ECAN	Attention and runtime allocator; also a causal gate carrier.
ACS / RAF	Candidate modular causal mechanism if it survives ablation.
Event log	Observational and replay substrate for influence estimation.
Reducer	Synchronization boundary for interventional summaries.
Worker graph	Discrete update system over causally factorized modules.

For each semantic implication rule:

```
R : C ^ G1 ==> G2
```

we maintain three conceptually distinct quantities, each answering a different question about the rule:

```
intensional strength:
    how much the properties or patterns of G1 tend to imply
    the properties or patterns of G2 in context C

causal influence:
    whether enabling, disabling, or perturbing R changes downstream
    answer reward, prediction error, hypothesis utility, story coherence,
    ACS formation, or other task measures

clarity / nonredundancy:
    whether R carries a direct useful causal signal, or merely duplicates
    a diffuse multi-hop path already present elsewhere
```

The first is essentially what PLN computes from data. The second can only be measured by running the chemistry under controlled perturbations. The third requires comparing alternative paths through the rule graph. PLN primarily supplies the first. ActPC-Chem discovers loops and recurrent useful structures involving many such rules. Causal coding estimates the second and third and feeds them back into ECAN, scheduling, rule evolution, and ACS selection. The result is a control loop in which the chemistry's resources flow toward rules that are plausible, useful, and non-redundant.

3.3 ActPC-Chem as a discrete causal-coding system

To make the connection to causal-coding theory explicit, it helps to state the chemistry in operator form. Let the semantic chemistry state be:

```
S =
  semantic graphs
  PLN implication rules and truth values
  rule weights and causal gates
  ECAN atom-level and worker-level attention values
  ACS macro-atoms
  context-chamber states
  event logs
  worker budgets
  reward models and evaluators
```

A context `c` induces an update operator:

$$U_c : S \to S.$$

That is, running the chemistry for some interval inside context `c` yields a new state. Different contexts induce different operators, each mostly touching the parts of the state most relevant to that context. Examples include:

```
U_biomed_causal_hypothesis:
    run biomedical semantic chambers;
    update implication rule weights and gates;
    reward useful causal hypotheses;
    create or revise ACS macro-atoms.

U_fiction_memory_world:
    run fictional-world semantic chambers;
    update world-rule implications and gates;
    reward coherent surprising plot events;
    create or revise story-generating ACSs.
```

The causal-coding question is:

```
Do these context updates touch mostly the modules that causally matter
for their contexts?
```

If they do, then for two mostly independent contexts `c1` and `c2` one expects:

$$U_{c_1}(U_{c_2}(S)) \approx U_{c_2}(U_{c_1}(S)).$$

That is, the order in which the system experiences the two contexts should not much matter, because each leaves the other's relevant modules mostly untouched. If they do not commute approximately, then the order of experience matters strongly and the system is vulnerable to semantic drift, forgetting, and habit traps. This is the discrete-symbolic analog of the causal-continual-learning theorem: causal modularity plus small commutators makes sequential learning close to joint learning, and error is controlled by a sparse graph of genuinely confusing context pairs [4]. In words: the more the system's internal modules align with the world's causal structure, the more its learning becomes stable and additive rather than fragile and overwriting.

3.4 Causal modules in semantic graph chemistry

What plays the role of a "module" in this setting? In a neural net it might be a head or a channel. In our setting it must be something native to symbolic-semantic structure. In this setting the modules are symbolic-semantic structures such as:

```
semantic relation families:
    Cause, Prevents, Enables, PartOf, GoalOf, EvidenceFor

domain modules:
    immune-metabolic, neurovascular, mitochondrial,
    legal-memory, kinship, trade, geography, oath-law

rule families:
    causal-abduction rules
    paraphrase-collapse rules
    answer-projection rules
    mechanism-completion rules
    analogy-bridge rules

ACS macro-atoms:
    reusable semantic reasoning loops

context chambers:
    biomedical-causal chamber
    fiction-world-law chamber
    why-question chamber
    counterfactual-question chamber
```

A context `C` should activate a relatively small support set of these modules:

$$Support(C) = \{\text{modules causally relevant to } C\}.$$

The smallness matters. A context that activates almost everything is no context at all; a context that activates almost nothing is useless. The goal is a meaningful intermediate, where each context picks out the handful of modules that genuinely bear on it.

For example, a biomedical post-viral cognitive-fatigue context might support:

```
immune activation
endothelial dysfunction
microvascular perfusion
mitochondrial stress
neuroinflammation
exertion response / PEM-like dynamics
trial endpoint design
```

whereas a fictional memory-currency context might support:

```
memory transfer
legal oath validity
taxation
collective geography
archive politics
identity boundaries
```

These two support sets barely overlap, which is exactly what we want: a night spent reasoning about post-viral fatigue should not silently rewrite the system's understanding of fictional taxation. Causal coding tries to ensure that updates from one chamber do not sloppily rewrite unrelated semantic modules. When two contexts do interfere, the interference should be explicit and interpretable rather than a side effect of ungated global rule reinforcement.

3.5 Causal gates on semantic PLN rules

To make causal coding actionable, each rule needs an explicit record of its causal status. A semantic rule record can be extended as follows:

```
(rule R)
(rule-lhs R G1)
(rule-rhs R G2)
(rule-context R C)

(rule-tv R intensional-implication s c)

(rule-do-influence R target answer-reward delta conf)
(rule-do-influence R target prediction-error delta conf)
(rule-do-influence R target acs-yield delta conf)
(rule-do-influence R target novelty-yield delta conf)

(rule-clarity R q)
(rule-redundancy R rho)
(rule-causal-module R M)
(rule-causal-gate R C g)
```

The `do-influence` entries are the central additions. Each one estimates how much enabling or disabling the rule changes a particular downstream outcome. Because these are interventional rather than correlational, they can be measured directly by paired replay:

```
same snapshot
same seed
same chamber budget
same question or task context

condition A:
    rule R enabled, boosted, or made hot

condition B:
    rule R disabled, masked, or made cold

measure:
    answer reward difference
    prediction-error difference
    downstream ACS formation difference
    hypothesis novelty/usefulness difference
    story coherence/surprise difference
```

This is the chemistry analog of an A/B test, run not on users but on the inference ecology itself. A simple causal gate can then combine the PLN and causal-coding quantities into a single firing propensity:

$$Gate_C(R) = \sigma(a \operatorname{logit}(s_R) + bc_R + d\, DoInfluence_C(R) + e\, NoveltyYield_C(R) \\ - f\, Redundancy_C(R) - g\, CpuCost(R)).$$

The gate then affects how the rule participates in the chemistry:

```
(pool-hot C R)
rule sampling probability
ECAN wage and rent
MOSES mutation prior
ACS dividends
worker scheduling priority
```

In this way, causal coding becomes part of the chemistry rather than an external audit applied afterward. The system's operating dynamics carry its causal accountability rather than treating it as a post-hoc check.

3.6 Causal coding as soft double-counting control

We can now revisit the earlier discussion of double-counting. The earlier design principle was:

```
Do not require exact global evidence bookkeeping.
```

Causal coding refines this to:

```
Do not require exact global evidence bookkeeping.
Do require local interventional accountability.
```

The shift is from "every piece of evidence is counted exactly once" to "every active loop earns its keep". So the system may tolerate multiple paths:

```
G1 -> G2
G1 -> H -> G2
G1 -> K -> L -> G2
```

but it asks of each:

```
Does this path add unique causal utility?
Does removing it change the outcome?
Does it merely inflate confidence through redundant correlation?
Does it help in one context but damage another?
```

This is a better fit to semantic chemistry than exact global de-duplication. Exact de-duplication tries to preserve a clean probabilistic interpretation at the cost of crippling the chemistry's exploratory dynamics. Soft, intervention-based accountability lets the chemistry remain wild while still being self-critical. The loopy graph remains loopy, but redundant loops pay rent, lose clarity, or fail ablation tests. Useful loops become causal semantic ACSs.

3.7 Route A, Route B, and the dependence tradeoff

The independence-coupling tradeoff gives another way to state the same design choice [5]. The framework distinguishes two ways of coping with statistical dependence in a learner: either restructure the representation so that dependence largely vanishes (Route A), or model the residual dependence explicitly (Route B). Most practical systems do some of each. In semantic graph chemistry:

```
Route A: independence-enforcing transforms
    normalize semantic graphs
    separate contexts
    factor causal modules
    prune diffuse redundant paths
    keep chamber updates local
    gate noncausal rules

Route B: explicit dependency modeling
    add PLN implication links
    add causal and associational edges
    add semantic rewrite rules
    form ACSs
    learn bridge rules between modules
```

Causal coding is the policy for balancing the two:

```
Use explicit semantic dependencies only where they carry causal signal.
Otherwise refactor, normalize, gate, split, prune, or tax the module.
```

In other words, when a dependency is doing real work, model it explicitly; when it is not, restructure so that it never needed to be modeled in the first place. FUSE-NF belongs largely to Route A: it makes the molecule space more stable, canonical, and modular. PLN implication links and ACS couplings belong largely to Route B: they explicitly model residual dependencies that matter. Semantic graph algorithmic chemistry needs both, and causal coding helps keep the system near a useful Pareto frontier between them.

3.8 Why FUSE-NF helps causal coding

Causal coding depends on measuring interventions over stable modules. This places a hard constraint on the representation: if the same underlying meaning shows up in many surface forms, then "intervening" on it is ill-defined, because the system cannot tell which surface form to perturb. Without semantic normalization, the system may treat these as unrelated rules:

```
rain caused the drive
bad weather made her go
storm forced the trip
wet roads motivated travel
```

Each of these expresses essentially the same causal claim, but to a naive graph chemistry they look like four different rules. Intervention tests on any one of them will give noisy or misleading results, because the others continue to fire and provide the same effect. With FUSE-NF-like consolidation, they become nearby or partially merged semantic molecules. Then causal coding can ask whether the underlying module:

```
WeatherEvent -> TravelMotivation
```

has real interventional influence in a why-question context. Now the intervention is well-defined, because the module is a single coherent target. Thus FUSE-NF is not merely a convenience for parsing. It supplies the stable module geometry that causal coding needs.

3.9 Causal semantic ACSs

Putting these ideas together, we can refine the central concept of an active autocatalytic set. A normal semantic ACS is a self-sustaining loop of rules:

```
rules produce graph motifs
graph motifs activate more rules
the loop generates useful outputs
```

This is already a meaningful object, but it does not yet guarantee that the loop is causally important. A causal semantic ACS additionally passes intervention tests:

```
ablating the ACS changes outputs
activating the ACS changes outputs
its members have nontrivial do-influence
its utility is not fully reproduced by redundant paths
its update effects are localized to an interpretable module
```

So we distinguish:

```
Active useful ACS:
    self-sustaining + rewarded

Causal active useful ACS:
    self-sustaining + rewarded + intervention-relevant
```

This is the object most relevant for scientific discovery and robust creative reasoning. It is not enough that a loop is lively. It must matter. The distinction is the difference between a recurring pattern that happens to keep firing and a recurring pattern that the system genuinely depends on.

3.10 The commutator diagnostic

The commutator condition stated earlier can be turned into a practical diagnostic. For two contexts `c1` and `c2`, periodically measure:

$$\Delta(c_1, c_2; S) = D\big(U_{c_1}(U_{c_2}(S)), U_{c_2}(U_{c_1}(S))\big),$$

where `D` may combine many notions of distance between system states:

```
answer-reward difference
semantic graph distance
rule-weight distance
causal-gate distance
ECAN attention distance
ACS archive difference
worker-budget difference
computational-cost difference
```

If `Delta` is small, the contexts are compatible or well-factorized: doing `c1` then `c2` produces approximately the same system as doing `c2` then `c1`, and the system can safely move between them. If it is large, they are a confusing pair: the order of exposure shapes the resulting system in ways that leak across context boundaries. The system can respond by:

```
splitting a chamber
creating a context-specific rule copy
adding an explicit bridge rule
```

```
increasing rent on overgeneral rules
lowering transfer between contexts
creating an ACS macro-atom with a guarded context
```

This is the semantic-chemistry analog of keeping context-conditioned update operators approximately commuting. It turns an abstract theorem about continual learning into a concrete operational signal that the chemistry can monitor and respond to.

3.11 Practical algorithm sketch

Bringing the whole picture together, a causal-coded semantic chemistry can be implemented as follows:

```
1. Parse and normalize text or domain data into FUSE-NF-like semantic graphs.

2. Mine PLN intensional implications:
       C ^ G1 ==> G2
   with strength and confidence from corpus statistics, graph pattern overlap,
   paraphrase data, entailment data, QA traces, and domain knowledge bases.

3. Partition rules and graph motifs into tentative causal modules:
       domain modules, relation modules, question-type modules,
       ACS candidates, context chambers.

4. Run semantic ActPC-Chem:
       MM2 chambers apply hot semantic rewrite rules;
       ECAN updates atom-level and worker-level attention;
       MOSES/CENF evolves rule structures;
       ACS detection finds active useful loops;
       QA/hypothesis/story reward supplies external selection.

5. Estimate causal influence:
       paired replay;
       rule ablation;
       rule activation;
       edge masking;
       do-style graph perturbation;
       chamber-level intervention;
       ACS ablation.

6. Update causal gates:
       strengthen rules with high intensional strength and high do-influence;
       weaken rules with high correlation but low intervention value;
       prune or tax redundant diffuse paths;
       assign worker runtime to high-yield modules.

7. Track cross-context commutators:
       if updates commute, permit transfer;
       if updates do not commute, split modules or add explicit bridge rules.
```

Each step uses the products of the previous ones, but the loop is not strictly sequential: in a deployed system steps 4–7 run continuously and feed each other, while steps 1–3 happen episodically as new data arrives.

3.12 What causal coding adds

It is worth stating plainly what the causal-coding layer buys. Causal coding gives semantic graph chemistry four important benefits:

```
1. It distinguishes semantic plausibility from causal usefulness.

2. It lets loopy PLN inference remain loopy while damping redundant
   non-interventional loops.

3. It turns ACSs into testable causal modules rather than merely
   self-reinforcing motifs.

4. It gives a continual-learning analysis: semantic chambers can adapt
   over time without catastrophically damaging one another, provided their
   updates are causally modular and their commutators stay small.
```

Thus the combined framework is:

```
Semantic PLN implications:
    weighted reaction rules over normalized semantic graphs

ActPC-Chem:
    creative, adaptive, reward-guided loopy inference ecology

Causal coding:
    interventional modularity discipline over the ecology

Result:
    a semantic reasoning chemistry that can be creative without collapsing
    into uncontrolled correlation loops or catastrophic cross-context
    interference
```

4 Natural language and semantic graphs

4.1 Why semantic normal form matters

Up to this point the discussion has been domain-agnostic. The first domain we treat in detail is natural language, because it is both the most accessible and the one where the cost of skipping normalization is most obvious.

Natural language has massive surface variability. The same event, relation, or causal pattern can be expressed in dozens of phrasings, with different word orders, different choices of voice, different idioms, and different levels of explicitness. Without a semantic normal form, a graph chemistry wastes energy rediscovering that many different phrases express nearly the same event, relation, role pattern, or causal motif. Worse, it cannot effectively combine evidence across these phrasings, because each one looks like its own isolated rule. FUSE-NF addresses this by constructing a fuzzy semantic normal form for natural language meaning. It starts from parallel structured interpretations, cross-links them, applies data-driven consolidation rules, adds background PLN inferences, labels causal and correlational edges, and optionally encodes the result with a graph transformer [3].

In this architecture, FUSE-NF is not merely preprocessing. It is a molecule-shaping discipline. It determines which semantic fragments become stable enough to support a useful chemistry. A useful image is that FUSE-NF determines the "allowed shapes" of molecules in our chemistry; only after such shapes have been fixed does it become meaningful to ask which reactions are common and which are rare.

```
sentence
    -> semantic parse(s)
    -> SENF / FUSE-NF graph
    -> semantic molecules
    -> PLN-weighted graph rewrite rules
    -> context chambers
    -> answer graph or story event graph
    -> natural language realization
```

The fuzzy part is especially important. In ordinary crisp symbolic inference, paraphrase fuzziness is a nuisance: if two graphs are not literally identical, classical unification fails. In semantic graph chemistry, paraphrase families are a substrate from which useful rewrite ecologies can emerge. Soft-edged molecules can react softly, and reactions that would be illegal in a strict logical setting become useful when interpreted as weighted, contextual transformations.

4.2 A FUSE-NF-like graph as a molecule

To make this concrete, consider a sentence such as:

```
After finishing her homework, Maria quickly drove to the store with John
because it was raining heavily.
```

A FUSE-NF-like graph may consolidate the syntax-derived and logic-derived structures into a compact graph such as:

```
Drive
|--- Participants : {Maria, John}
|--- Destination  : store
|--- Manner       : quickly
|--- Temporal     : AfterFinishHomework

Rain
|--- Manner       : heavily

Cause
|--- CauseOf      : Rain
|--- EffectOf     : Drive
```

This is not the only way to draw the graph; a parser might emit several competing structures and the normalization step picks or averages among them. Background PLN inference may then add what was not stated but is strongly implied:

```
Completed(Homework)     confidence 0.95
Motion(DriveEvent)      confidence 0.90
MaybeWet(store)         confidence 0.85
```

Causal and correlational edges may then distinguish what the surface sentence does not separate:

```
Rain -> Drive           causal or explanatory
FinishHomework -> Drive temporal/precondition-like
Participants <-> quickly correlational or stylistic
```

This graph, or a subgraph of it, can be a molecule in the chemistry. The labels distinguish what kind of edge each one is, which is exactly what later causal-coding queries will need.

4.3 Rules over normalized semantic graphs

A semantic reaction rule can now operate over these normalized fragments. Because the molecules are clean and consistent, the rules can be written generically and matched widely:

```
(rule R_weather_reason_for_travel)

(context R_weather_reason_for_travel C_weather_why)

(lhs R_weather_reason_for_travel
  (, (Rain ?e_rain)
     (Drive ?e_drive)
     (Agent ?e_drive ?person)
     (Destination ?e_drive ?place)
     (Cause ?e_rain ?e_drive)))

(rhs R_weather_reason_for_travel
  (, (ReasonFor ?e_drive ?e_rain)
     (WeatherMotivatedTravel ?e_drive)
     (MaybeWet ?place)))

(tv R_weather_reason_for_travel
  (IntensionalImplication lhs rhs 0.78 0.61))
```

This rule encodes a specific kind of inferential move: when a rain event is reported as causing a drive event, infer that the drive is weather-motivated and that the destination may be wet. The behavior of this rule depends sharply on the active context. In a chamber contextualized to a why-question about travel, this rule becomes hot. In a general travel-description chamber, it may be warm. In a purely geographical chamber, it may be cold. A single rule's relevance varies by chamber, and that variation is precisely what makes contexts useful.

4.4 Contextual semantic chambers

The contextual rule form:

```
C ^ G1 ==> G2
```

is central, and the kinds of `C` we allow strongly shape what the chemistry can do. The context `C` can represent:

```
semantic context:
    medical, legal, physics, social, fictional, mathematical

discourse context:
    current paragraph, speaker, time, presuppositions

question context:
    why-question, where-question, analogy-question, counterfactual-question

reasoning style:
    cautious, creative, explanatory, abductive, causal, analogical
```

```
source context:
    document, agent, corpus, modality, confidence regime
```

Different combinations pick out different reasoning regimes. A chamber can therefore be specialized:

```
CH_medical_why:
    context = medical ^ causal ^ why-question

CH_story_abduction:
    context = narrative ^ character-intention ^ abductive

CH_science_explanation:
    context = physics ^ causal ^ explanation

CH_paraphrase_bridge:
    context = semantic-normalization ^ paraphrase
```

Each chamber pulls on a different population of rules and tends to form different ACSs. This also fits the execution model in the long companion paper: each chamber can be run as a worker with its own hot pool, seed, budget, and ordered event log [1].

4.5 Question answering as external reward

Inference is most useful when it has something it is trying to do. In the natural language setting, a particularly clean source of selection pressure is question answering: the chemistry has to produce an answer graph that satisfies a question skeleton, and the quality of that answer becomes its reward.

A natural-language question is mapped into a question graph and an answer skeleton. For example:

```
question:
    Why did Maria drive to the store?

question graph:
    (Question Q1)
    (QuestionType Q1 why)
    (TargetEvent Q1 ?drive)
    (Drive ?drive)
    (Agent ?drive Maria)
    (Destination ?drive Store)

answer skeleton:
    (Answer A1)
    (Answers A1 Q1)
    (ReasonFor ?drive ?cause)
```

The question graph encodes what the question is about; the answer skeleton encodes the shape that any acceptable answer must take. The semantic chemistry runs in one or more contextual chambers until candidate answer graphs appear:

```
candidate answer graph:
    (Answer A1)
    (Answers A1 Q1)
```

```
    (ReasonFor DriveEvent RainEvent)
    (Rain RainEvent)
    (Support A1 LOG123)
```

External reward can be computed from several sources, each capturing a different facet of what makes an answer good:

```
task reward:
    did the answer match a known answer or evaluator judgement?

semantic reward:
    does the answer graph satisfy the question skeleton?

PLN reward:
    is the answer supported by high-confidence implication paths?

causal reward:
    for why/how questions, does the answer use causal edges?

compression reward:
    is the justification compact?

novelty reward:
    did the answer require a nontrivial semantic bridge?
```

Combining several of these is important: optimizing for any one alone tends to produce degenerate behavior (e.g. purely novel answers that are wrong, or purely correct answers that recycle the same trivial path). Thus the external task supplies selection pressure, while the chemistry supplies creative inference control.

4.6 Semantic ACSs

We can now describe what an active useful ACS looks like in the linguistic setting. An active useful ACS in this domain is not just a set of graph fragments. It is a self-sustaining loop of semantic transformations that repeatedly helps generate useful answers, explanations, hypotheses, or story events.

A semantic ACS might include:

```
a question-type detector
+ a causal-alignment rule
+ a role-completion rule
+ a paraphrase-collapse rule
+ a PLN implication bridge
+ an answer-projection rule
+ a surface-realization pattern
```

Together these form a closed cycle in which detecting a question type brings up causal-alignment, which brings up role-completion, which eventually feeds back into the next question detection. It is active because its members repeatedly trigger each other in actual event logs. It is useful because ablation shows that removing the loop reduces task reward, explanation quality, novelty, or transfer. In effect, the system has discovered a reusable inference strategy and packaged it as a recurrent network of rules.

5 Learning semantic reaction rules

The chemistry needs a steady supply of rules, and these can come from many sources. No single source is sufficient; in practice a deployed system blends them. Semantic reaction rules can be learned or seeded from many sources.

Source	Rule-learning role
Paraphrase pairs	Learn bidirectional or asymmetric mappings between near-equivalent semantic graphs.
Entailment data	Learn premise-graph to hypothesis-graph implications.
QA traces	Credit transformations that led to rewarded answer graphs.
PLN mining	Estimate intensional implication between variable-bearing subgraphs from property overlap and contextual coactivation.
Causal data	Learn cause-pattern to effect-pattern rules, or context plus intervention to outcome rules.
Analogy	Learn mappings between graph motifs in different domains.
FUSE-NF consolidation	Collapse recurring graph fragments into meta-nodes and use those meta-nodes as rule atoms.
Human seed knowledge	Supply initial rules, contexts, reward hints, and safety constraints.

The same rule may be evaluated along several axes simultaneously, since different downstream uses care about different things. A rule may have multiple scores:

```
(rule-tv R intensional-implication 0.78 0.61)
(rule-tv R causal-support          0.45 0.30)
(rule-tv R paraphrase-support      0.86 0.74)
(rule-tv R qa-utility              0.63 0.52)
(rule-tv R novelty-yield           0.18 0.40)
(rule-cost R cpu-ms-expected       3.5)
```

The chemistry need not collapse all of these into one static weight. ECAN, worker-level scheduling, task reward, ACS dividends, and ablation can all use different views of the rule. A rule that is weak as a pure implication but strong as a paraphrase bridge, for example, may still earn its keep when used by a paraphrase-collapse worker.

6 Conceptual example 1: biomedical research creativity

6.1 Caveat

The following biomedical example is a conceptual illustration of hypothesis generation. It is not medical advice, not a diagnosis, not a treatment recommendation, and not a claim that the generated hypothesis is established. The point is to show how a semantic graph chemistry could combine weak, contextual biomedical implications into creative, testable research programs.

6.2 Domain: post-viral cognitive fatigue

The example domain is chosen because it is genuinely hard: the biology is multi-system, the symptom patterns vary across patients, and no single mechanism explains every case. This is exactly the kind of domain where loop-finding chemistry could plausibly outperform single-pathway reasoning.

Take the motivating domain to be post-viral cognitive fatigue, using long-COVID-like brain fog as an illustrative example. Suppose a biomedical corpus has been parsed into semantic graphs such as:

```
G_a:
  (PostViralState ?p)
  (PersistentImmuneActivation ?p)
  (InflammatoryMarkerElevated ?p ?m)
  (Symptom ?p CognitiveFatigue)

G_b:
  (EndothelialDysfunction ?p)
  (MicrovascularPerfusionReduced ?p ?tissue)
  (ExerciseChallenge ?p)
  (PostExertionalMalaise ?p)

G_c:
  (MitochondrialStress ?cell)
  (ROSIncreased ?cell)
  (EnergyProductionReduced ?cell)

G_d:
  (BloodBrainBarrierVulnerable ?p)
  (MicroglialPriming ?p)
  (SynapticPlasticityReduced ?p)
  (Symptom ?p BrainFog)

G_e:
  (GutDysbiosis ?p)
  (ImmuneToneShifted ?p)
  (TregBalanceAltered ?p)
```

The corpus does not need all these propositions to be certain. In fact they should not be: real biomedical evidence is a mixture of replicated results, single-study claims, mechanistic hypotheses, and conflicting interpretations. PLN truth values, provenance, and source-context annotations allow the chemistry to reason under uncertainty rather than collapse it prematurely.

6.3 Example reaction rules

From data and prior knowledge, the system learns or receives rules such as:

```
R1:
  C_postviral_neuroimmune ^
  (PersistentImmuneActivation ?p)
  ==>
  (MicroglialPriming ?p)

  tv: IntensionalImplication 0.68 0.42
```

```
R2:
  C_neurovascular ^
  (EndothelialDysfunction ?p)
  (MicrovascularPerfusionReduced ?p Brain)
  ==>
```

```
  (CognitiveEnergyBottleneck ?p)

  tv: IntensionalImplication 0.61 0.39
```

```
R3:
  C_metabolic ^
  (CognitiveEnergyBottleneck ?p)
  (MitochondrialStress ?cell)
  ==>
  (ExertionAmplifiesSymptoms ?p)

  tv: IntensionalImplication 0.58 0.36
```

```
R4:
  C_gut_immune ^
  (GutDysbiosis ?p)
  (ImmuneToneShifted ?p)
  ==>
  (PersistentImmuneActivation ?p)

  tv: IntensionalImplication 0.55 0.34
```

None of these rules is a dramatic discovery by itself. Each one is the sort of moderate-strength, moderate-confidence claim that one might extract from a careful reading of the literature. The creative behavior appears when a reaction chamber lets them interact repeatedly in context. Individually they are modest; collectively, the loops they form become hypotheses with structure.

6.4 Context chamber

To focus the chemistry, we set up a chamber whose context activates exactly the modules this domain needs:

```
CH_postviral_brainfog:
  context:
    postviral ^ cognitive-fatigue ^ PEM ^ neurovascular ^ immune-metabolic

  hot molecules:
    graphs about brain fog, PEM, endothelial dysfunction, microglia,
    mitochondrial stress, dysbiosis, autonomic dysfunction, candidate
    biomarkers, trial endpoints, and intervention classes

  external reward:
    produce a causal or therapeutic research hypothesis that:
      explains multiple symptom clusters;
      gives subtype markers;
      suggests testable intervention classes;
      avoids unsupported cure claims;
      generates falsifiable predictions.
```

The reward function is deliberately demanding. It penalizes both empty abstraction (a hypothesis that explains nothing specifically) and overreach (a hypothesis that pretends to therapeutic certainty). The chemistry has to find configurations that sit in the narrow useful zone between these two failure modes.

6.5 A creative ACS-like loop

A conventional inference engine might produce isolated one-hop hypotheses:

```
immune activation -> neuroinflammation -> brain fog
mitochondrial dysfunction -> fatigue
vascular dysfunction -> poor perfusion
```

Each of these is an honest local inference, and each is too narrow on its own to constitute a research program. The semantic chemistry may instead find a loop like:

```
EndothelialDysfunction
  -> MicrovascularPerfusionReduced
  -> CognitiveEnergyBottleneck
  -> ExertionAmplifiesSymptoms
  -> AutonomicCompensation
  -> BBBVulnerability
  -> MicroglialPriming
  -> NeuroimmuneFeedback
  -> EndothelialDysfunction
```

This is not a proof. It is a circular pattern in which each step makes the next step more plausible, and the closing edge ties the cycle back to its start. It becomes interesting when the loop is autocatalytic in the semantic chemistry: inferring one part of the loop makes other rules applicable, and successful answer traces repeatedly reactivate the same motif. The loop is not a proof. It is a self-sustaining semantic reaction network.

The ACS detector might label it:

```
ACS_42:
  name: neurovascular-immune-energy feedback loop

  members:
    R_endothelium_to_energy
    R_energy_to_PEM
    R_PEM_to_autonomic_compensation
    R_BBB_to_microglia
    R_microglia_to_neuroimmune_feedback
    R_immune_to_endothelium

  usefulness:
    explains brain fog + PEM + autonomic-like symptom patterns
    predicts subtype markers
    suggests multi-site intervention tests
```

The naming is significant. Once the chemistry has identified a recurrent useful loop, that loop becomes a single named object with which other parts of the system can reason.

6.6 Generated research hypothesis

A candidate output, expressed in something close to natural language, could be:

```
Hypothesis H42:

In a subtype of post-viral cognitive fatigue, brain fog is not primarily a
single-site CNS disorder or a single-site mitochondrial disorder.  It is a
```

```
neurovascular-immune-energy feedback loop.

The key causal bottleneck is the coupling between:
  microvascular perfusion instability,
  mitochondrial stress under exertion,
  blood-brain-barrier or neuroimmune vulnerability,
  and autonomic compensation.

Therapeutic research should test staged or combined intervention classes that
first stabilize the neurovascular/autonomic context and then improve cellular
energy handling, rather than testing isolated cognitive training or isolated
mitochondrial support in unstratified patients.
```

The candidate answer graph backs this up with explicit subtype markers and trial-design constraints:

```
(subtype H42 S_neurovascular_energy)
(marker S_neurovascular_energy OrthostaticIntoleranceLikeFeature)
(marker S_neurovascular_energy ExertionTriggeredCognitiveDrop)
(marker S_neurovascular_energy InflammatoryOrStressMarkerShift)
(marker S_neurovascular_energy PerfusionOrEndothelialMarkerShift)

(test H42 TrialDesign)
(requires TrialDesign BaselinePhenotyping)
(requires TrialDesign LongitudinalPEMMeasurement)
(requires TrialDesign PlaceboOrControlArm)
(requires TrialDesign SubgroupAnalysis)
```

This output is not merely "try therapy X." It is a structured research-program graph:

```
causal loop hypothesis
+ subtype definition
+ marker set
+ trial design constraint
+ falsifiable predictions
```

The shape matters. A research-program graph is the kind of object a research group could actually act on, refine, and either confirm or falsify.

6.7 Causal-coding analysis of the biomedical loop

We can now apply the three-layer analysis to this hypothesis. A pure semantic-PLN view of the loop says that the implication links have moderate intensional strength and cohere as an explanation graph. The algorithmic-chemistry view says that the loop repeatedly helps produce rewarded biomedical hypotheses and therefore becomes an active useful ACS. The causal-coding view adds a more demanding question: does the loop matter under intervention? Is it doing real work in the system, or is it just an attractive narrative that the chemistry happens to settle into?

For the loop:

```
PersistentImmuneActivation
  -> EndothelialDysfunction
  -> MicrovascularPerfusionReduced
  -> CognitiveEnergyBottleneck
  -> PostExertionalSymptomAmplification
```

```
  -> NeuroimmuneFeedback
  -> PersistentImmuneActivation
```

paired replay and ablation can ask several specific operational questions:

```
If the neurovascular-energy loop is ablated,
    do hypothesis rewards drop?

If only the immune-to-microglia rule is disabled,
    does the system still produce equally useful explanations by another path?

If the module is transferred to a related post-viral fatigue context,
    does it help or interfere?

Does the module generate predictions under intervention,
    not merely retrospective semantic coherence?
```

If the loop passes these tests, the hypothesis is therefore upgraded from a plausible explanation to a causally testable module:

```
The useful causal unit may be a feedback loop, not a node.
The intervention target may be loop stabilization, not one pathway.
The trial design should phenotype patients according to predicted bottlenecks.
```

This reframing is itself a contribution. It shifts both the scientific target (loops, not nodes) and the operational target (loop stabilization, not single-pathway treatment). Causal coding then gates the chemistry toward rules and ACSs that survive these intervention-like tests.

6.8 Where the creativity enters

It is worth being explicit about where the creative work happens. The creative step is not any single rewrite. No single rule above is novel on its own. It is the loop becoming a useful attractor. The reaction chamber repeatedly combines weak, domain-specific implications:

```
vascular biology rule
+ immune feedback rule
+ mitochondrial stress rule
+ exertion/PEM rule
+ neuroinflammation rule
+ trial-design rule
```

A summary system might list these as separate mechanisms. A symbolic inference system might produce isolated chains. The algorithmic chemistry can instead discover a semantic ecology:

```
The target is not one node.
The target is a feedback loop.
Patient selection is part of the hypothesis.
Endpoint design is part of the hypothesis.
```

That is a specific kind of biomedical creativity: the invention of a structured, testable, multi-graph research program from interacting uncertain semantic motifs. It resembles, at a high level, what a good research clinician might do when synthesizing many overlapping studies into a coherent program of investigation.

7 Conceptual example 2: fiction generation creativity

7.1 Goal

The fiction-generation example illustrates a different kind of creativity. It is in some ways the dual of the biomedical case: there the goal was to find loops that survive intervention tests over real biology, and here the goal is to discover events that follow inevitably from invented but locally consistent world rules. The goal is not to retrieve or imitate a familiar trope. The goal is to let world rules chemically react until they generate events that are surprising, coherent, character-relevant, and native to the fictional world.

7.2 World premise

The setup is to describe a world in terms of a small number of strong, unusual rules and let the chemistry discover the consequences. Suppose the world context is:

```
C_memory_archipelago:

In the Archipelago of Veyr, memories are transferable objects.
A memory can be stored in glass, traded, taxed, stolen, forged, or burned.
Collective memory partly stabilizes geography: a harbor remains navigable
because enough people remember its correct shape.
Oaths are legally binding only while at least one party remembers making them.
```

The context is a strong catalyst. It forces ordinary semantic rules about economics, identity, law, geography, family, memory, and magic to react through a shared premise:

```
memory is transferable value
memory is personal identity
memory is legal evidence
memory stabilizes physical reality
```

Each of these collapses a distinction we usually take for granted. The interesting fiction will arise from following the consequences of these collapses in interaction.

7.3 Semantic graph molecules

The starting molecules are abstract enough to apply to many situations in this world:

```
G_currency:
  (Currency ?x)
  (Transferable ?x)
  (CanBeTaxed ?x)
  (CanBeStolen ?x)
  (CanInflate ?x)
  (MarketFormsAround ?x)

G_memory:
  (Memory ?m)
  (PartOfIdentity ?m ?person)
  (EvidenceForPastEvent ?m ?event)
  (SupportsRelationship ?m ?relationship)

G_oath:
```

```
  (Oath ?o)
  (Obligation ?o ?person)
  (ValidIfRemembered ?o)

G_geography:
  (Place ?p)
  (CollectivelyRemembered ?p)
  (StableIfRemembered ?p)
```

Learned or seeded intensional rules might include:

```
R_currency_theft:
  C_memory_archipelago ^
  (Currency ?x)
  (Transferable ?x)
  ==>
  (TheftMarketFormsAround ?x)
```

```
R_memory_identity:
  C_memory_archipelago ^
  (Memory ?m)
  (PartOfIdentity ?m ?person)
  (Transferred ?m ?person ?other)
  ==>
  (IdentityBoundaryBlurred ?person ?other)
```

```
R_oath_memory:
  C_memory_archipelago ^
  (Oath ?o)
  (ValidIfRemembered ?o)
  (MemoryOf ?o ?m)
  (Destroyed ?m)
  ==>
  (ObligationCollapses ?o)
```

```
R_place_memory:
  C_memory_archipelago ^
  (Place ?p)
  (StableIfRemembered ?p)
  (CollectiveMemoryDamaged ?p)
  ==>
  (GeographyBecomesUnstable ?p)
```

7.4 Ordinary continuation versus chemical creativity

To see what the chemistry adds, contrast it with surface-level continuation. A surface-level continuation might produce a familiar plot:

```
A thief steals the hero’s memories.
The hero must recover them.
A villain controls the memory market.
The hero exposes the villain.
```

This is coherent and recognizable, but close to trope recombination. The world's specific rules are barely doing any work; the same plot could appear in a generic fantasy setting with the labels changed.

The chemistry can instead form interacting loops that exploit the world's structure:

```
MemoryAsCurrency
  -> MemoryTaxation
  -> PoorPeopleSellAncestralMemories
  -> CollectiveMemoryWeakens
  -> GeographyDestabilizes
  -> TradeRoutesFail
  -> StateRaisesMemoryTaxes
  -> MemoryAsCurrency
```

and:

```
OathsRequireMemory
  -> PoliticiansEraseOathMemories
  -> LawsLoseBindingForce
  -> CourtsRequireMemoryArchives
  -> ArchivesBecomePoliticalWeapon
  -> RebelsAttackArchives
  -> OathsRequireMemory
```

A high-reward bridge combines the loops into something that could not have come from a single rule:

```
CollectiveMemoryStabilizesGeography
+
LegalOathsRequireMemory
+
ArchivesStoreTaxedMemories
==>
Destroying or redistributing a legal archive can physically move borders.
```

7.5 Generated occurrence

That bridge can then be turned into a concrete plot event. The system might propose this Act II event:

```
During a famine, the Queen forgives the memory tax for one night.
Citizens rush to reclaim memories they had sold to the Treasury.

But many of those memories were not merely private.
They contained the shared recollection of the treaty line between Veyr and
the neighboring sea-kingdom.

When the people take their memories back in different fragments, the border
is no longer collectively remembered in one consistent way.

At dawn, the harbor has moved.

Ships that were legally inside Veyr are now anchored in enemy water.
A prison island appears in the middle of the capital lagoon, because only
```

```
the prisoners remembered where it truly was.
The Queen has not lost a battle; she has lost the public memory that made
the kingdom's map true.
```

This is not a random surreal event. Every consequence is licensed by the world rules: memory is currency (so it can be taxed and reclaimed), memory is legal evidence (so treaties depend on it), and collective memory stabilizes geography (so its fragmentation moves borders). It follows from several world-rule graphs reacting:

```
memory is currency
memory is legal evidence
collective memory stabilizes geography
state taxation extracts memory
treaties and borders depend on remembered facts
```

7.6 Candidate event graph

The generated event can be represented internally as a graph the system can reason about further:

```
(event E_harbor_moves)

(cause E_harbor_moves MemoryTaxAmnesty)
(cause MemoryTaxAmnesty FaminePressure)
(cause MemoryTaxAmnesty PoliticalMercyGesture)

(enables MemoryTaxAmnesty MassMemoryWithdrawal)
(affects MassMemoryWithdrawal TreatyMemoryArchive)
(affects TreatyMemoryArchive CollectiveBorderMemory)

(rule-applied R_place_memory)
(rule-applied R_oath_memory)
(rule-applied R_currency_theft)

(result E_harbor_moves GeographyBecomesUnstable)
(result E_harbor_moves HarborMoves)
(result E_harbor_moves ShipsBecomeForeign)
(result E_harbor_moves PrisonIslandReappears)

(narrative-function E_harbor_moves Reversal)
(narrative-function E_harbor_moves WorldRuleDeepening)
(narrative-function E_harbor_moves MoralComplication)
```

The external reward function may score this event highly because it is simultaneously several good things at once:

```
surprising:
    the harbor moves because of a tax amnesty, not because of a battle

coherent:
    every consequence follows from established world rules

character-relevant:
    the Queen's compassionate act creates a geopolitical disaster
```

```
plot-generating:
    it creates war risk, prison escape, legal crisis, and moral ambiguity

non-mimetic:
    it is not just "memory thief steals hero's past"
```

7.7 Causal-coding analysis of the fictional occurrence

The same causal-coding lens applies in fiction, even though "intervention" here means something looser than in biomedical work. In a fictional world, causal coding treats world laws as intervention-bearing modules. A story event is more interesting when it is surprising because it follows from the world's causal structure, not merely because it is bizarre. The world's rules play the role of laws of nature; we can ask what would happen if they were perturbed. For the memory archipelago, relevant interventions include:

```
do(remove memory tax)
do(split collective border memory)
do(destroy oath archive)
do(restore private memories)
do(mask legal memory records)
```

The causal-coded chemistry asks:

```
Which consequences follow because of the world's laws?
Which consequences are merely genre associations?
Which event changes the future state space most coherently?
Which world-rule module is responsible for the plot twist?
```

The first question is the key one. An event is interesting when the chain leading to it would survive intervention testing within the world's logic. The harbor-moving tax amnesty has high causal clarity because the chain is compact and intervention-bearing:

```
tax policy
  -> mass memory withdrawal
  -> collective border-memory fragmentation
  -> geography instability
  -> harbor relocation
  -> geopolitical crisis
```

A causal-gate record for the event might include:

```
(event-causal-module E_harbor_moves M_collective_memory_geography)
(event-do-influence E_harbor_moves future-plot-branching 0.82 0.55)
(event-do-influence E_harbor_moves character-dilemma 0.74 0.51)
(event-clarity E_harbor_moves 0.79)
(event-redundancy E_harbor_moves 0.18)
```

Thus the event is not only novel. It is novel in a way that exposes and tests the causal grammar of the invented world. This is arguably what distinguishes good speculative fiction from arbitrary strangeness.

7.8 Where the creativity enters

As before, it pays to be clear about where the creative move happens. Again, the creative step is not one rule. It is the emergence of an active semantic ecology:

```
economics of memory
+ law of oath-memory
+ geography of collective memory
+ famine politics
+ archive vulnerability
= harbor-moving tax amnesty
```

This differs from direct text continuation because the system is not only asking:

```
What text usually follows this text?
```

It is asking:

```
What semantic reaction network becomes active in this world context?
Which loops generate coherent consequences?
Which consequences are high-novelty and high-reward under the story objective?
```

These are different questions. The first is about surface mimicry; the others are about the chemistry of meaning under invented constraints.

8 Common pattern in the two examples

The biomedical and fiction examples are outwardly different, but structurally similar. Recognizing the parallel reveals what is domain-independent about the framework.

In the biomedical case, creativity arises when the system proposes:

```
Do not treat the disorder as one causal node.
Treat it as a feedback loop with subtype-specific intervention points.
```

In the fiction case, creativity arises when the system proposes:

```
Do not treat world rules as decorative lore.
Let them chemically react until they produce plot events.
```

In both, the gain comes from moving from isolated facts or rules to self-supporting reaction networks. The same architecture supports both:

```
semantic graphs:
    molecules

PLN intensional implications:
    weighted reaction rules

contexts:
    reaction chambers

ECAN:
    attention and runtime budgeting

ACS detection:
    discovery of self-sustaining inference/story/hypothesis loops

external reward:
    biomedical usefulness or narrative power
```

9 Beyond natural language

Although natural language is a natural first domain, the method is not limited to language. The requirement is fairly minimal: a supply of variable-bearing subgraphs and enough data to estimate intensional implication links among them. Wherever those exist, the same chemistry can in principle run.

Possible non-NL domains include:

```
biomedical pathway graphs:
    pathway motif ==> phenotype motif
    drug effect context ^ pathway motif ==> outcome motif

robotic scene graphs:
    spatial-action motif ==> affordance motif
    failure context ^ action graph ==> repair graph

software graphs:
    code pattern ==> bug pattern
    optimization context ^ program graph ==> faster program graph

mathematical proof graphs:
    lemma motif ==> proof-state transformation
    domain context ^ algebraic pattern ==> theorem skeleton

social or economic graphs:
    incentive motif ==> behavior motif
    policy context ^ network pattern ==> emergent outcome pattern
```

In each of these, the molecules and reactions look domain-specific, but the underlying machinery is the same. The general rule form remains:

```
C ^ G1 ==> G2
```

with a PLN-weighted intensional implication truth value and an algorithmic chemistry that searches for useful, active, self-sustaining transformation loops.

10 Implementation sketch

10.1 Atom schemas

Implementation begins with deciding how to store the central objects. A semantic graph chemistry can use schemas such as:

```
(sem-graph G17)
(sem-graph-kind G17 fuse-nf)
(sem-node G17 N3)
(sem-edge G17 E9 CauseOf N1 N2)
(sem-edge-tv G17 E9 causal 0.72 0.48)

(sem-rule R44)
(rule-lhs R44 G_lhs_44)
(rule-rhs R44 G_rhs_44)
(rule-context R44 C_weather_why)
(rule-tv R44 intensional-implication 0.78 0.61)
```

```
(rule-cost R44 cpu-ms-expected 3.2)

(qa-task Q12)
(question-graph Q12 Gq12)
(answer-skeleton Q12 Gas12)
(candidate-answer Q12 A77)
(answer-graph A77 Ga77)
(answer-reward A77 0.84)

(semantic-chamber CH_sem_why)
(chamber-context CH_sem_why C_weather_why)
(chamber-hot-rule CH_sem_why R44)
```

These schemas keep graphs, rules, tasks, and chambers as first-class records that can be queried, indexed, and reasoned about by other parts of the chemistry.

Causal-coding extensions are added as additional records on the same objects, so existing tools see them when needed and ignore them otherwise:

```
(rule-do-influence R44 target answer-reward 0.23 0.40)
(rule-do-influence R44 target prediction-error -0.12 0.33)
(rule-clarity R44 0.71)
(rule-redundancy R44 0.19)
(rule-causal-module R44 M_weather_travel)
(rule-causal-gate R44 C_weather_why 0.82)

(causal-module M_weather_travel)
(causal-module-kind M_weather_travel semantic-relation-family)
(causal-module-member M_weather_travel R44)
(causal-module-support-context M_weather_travel C_weather_why)

(context-update U_weather_why)
(update-context U_weather_why C_weather_why)
(update-support-module U_weather_why M_weather_travel)
(update-commutator U_weather_why U_story_abduction 0.07)

(acs-causal-test ACS42 ablation reward-drop 0.31 0.44)
(acs-causal-test ACS42 activation reward-gain 0.27 0.39)
(acs-clarity ACS42 0.76)
(acs-redundancy ACS42 0.22)
```

New worker kinds may include:

```
semantic-chamber
semantic-rule-miner
fuse-nf-normalizer
qa-evaluator
answer-surface-realizer
semantic-acs-scan
semantic-replay-batch
fiction-world-chamber
biomedical-hypothesis-chamber
```

Each worker kind specializes in one part of the cycle, and the worker graph as a whole orchestrates them.

10.2 Minimal prototype

It is tempting to start with the most ambitious version of the system, but the dynamics of semantic chemistry are easier to study and debug at small scale. A minimal prototype should avoid open-domain ambition at first.

```
domain:
    short stories, toy biomedical corpus, or constrained scientific abstracts

inputs:
    1000 to 10000 parsed sentences or graph records
    100 to 500 QA pairs or evaluation tasks

rules:
    mined graph implications G1 ==> G2
    hand-seeded QA projection rules
    hand-seeded causal and role-completion rules
    paraphrase consolidation rules

chambers:
    one chamber per question type, domain context, or narrative world context

reward:
    answer graph matches target skeleton;
    explanation is compact;
    novelty and causal coherence are rewarded;
    unsupported or incoherent claims are penalized.
```

For biomedical research creativity, the first benchmark should be a hypothesis ranking and explanation task over a constrained corpus, not automatic clinical recommendation. For fiction, the first benchmark should be generation of world-rule-consistent events, not unrestricted novel writing. In both cases the goal at this stage is to verify that the chemistry produces non-trivial loops and that those loops earn their keep.

10.3 Evaluation

Evaluation should not be limited to task performance. The chemistry's internal structure is itself an object of study, and metrics about its shape are needed alongside metrics about its outputs. Evaluation should include both task performance and chemistry analysis.

```
Task metrics:
    answer accuracy
    evaluator score
    explanation compactness
    narrative interest
    hypothesis plausibility
    novelty under nearest-neighbor comparison

Chemistry metrics:
    number of active semantic ACSs
    ablation impact of each ACS
    recurrence of useful motifs across tasks
    transfer of semantic ACSs to new contexts
    reward per CPU millisecond
```

```
    degree of rule-loop redundancy and damping

Causal-coding metrics:
    do-influence of rules, modules, workers, and ACSs
    clarity and redundancy of semantic pathways
    cross-context commutator defects
    number and weight of confusing context pairs
    intervention robustness of answers, hypotheses, and story events
    forgetting or semantic drift after sequential context updates

Replay metrics:
    event-log reproducibility
    provenance quality
    ability to explain generated answer or event graph
```

The key scientific question is whether useful, active semantic ACSs appear and whether their removal reduces performance or creativity. Everything else is in service of that question.

11 Conclusion

Semantic Graph Algorithmic Chemistry combines three ideas:

```
PLN:
    uncertain, contextual, intensional implication between graph patterns

FUSE-NF-like semantic normalization:
    stable molecule shapes for natural language meaning

Algorithmic chemistry:
    adaptive, reward-guided, autocatalytic inference-control ecologies

Causal coding:
    interventional modularity, causal gates, clarity, and commutator control
```

Each ingredient addresses a piece that the others leave open: PLN gives weighted contextual rules but no inference-control discipline; FUSE-NF gives stable representations but no dynamics; algorithmic chemistry gives dynamics but no accountability for what those dynamics actually accomplish; causal coding gives that accountability.

The result is a framework in which semantic graph rules are not merely inference rules. They are reactive entities in a contextual chemistry. Some rules fire, some catalyze others, some form loops, some become ACSs, some are rewarded by question-answering or story-generation success, and some are forgotten or taxed away by ECAN. Causal coding further asks which of these rules, loops, and modules have interventional influence. A loop that is lively but redundant loses clarity; a loop that changes downstream outcomes under ablation becomes a causal semantic ACS.

The most important conceptual move is to make creativity emerge from reaction-network structure, rather than locating it in any single rule or proof. In biomedical research, this can mean generating feedback-loop hypotheses and subtype-specific research programs rather than single-node causal guesses. In fiction, it can mean generating events that arise from interactions among world rules rather than copying familiar tropes. In both cases, the system is not merely continuing text or chaining proofs. It is cultivating semantic reaction ecologies. Causal coding gives those ecologies a discipline: plausible semantic transformations are allowed to interact freely, but the system continually estimates which interactions matter under intervention, which contexts commute, and which modules should be split, gated, taxed, or promoted. The combination is intended to be both wilder and more accountable than conventional inference: wilder because the chemistry tolerates loops, redundancy, and exploration that classical inference forbids; more accountable because every active loop must show that it earns its keep under perturbation.

References

[1] Ben Goertzel and various LLMs. *ActPC-Chem on MORK: Architecture, Design, and Implementation Reference*. Draft companion manuscript, 2026.

[2] Ben Goertzel, Matthew Ikle, Izabela Freire Goertzel, and Ari Heljakka. *Probabilistic Logic Networks: A New Conceptual, Mathematical and Computational Framework for Uncertain Inference*. Draft book manuscript, June 27, 2008.

[3] Ben Goertzel. *FUSE-NF: Constructing an Elegant Normal Form for Natural Language Semantics*. Concept note, May 26, 2025.

[4] Ben Goertzel. *Causal Coding and the General Causal-Continual-Learning Theorem*. Draft manuscript, December 1, 2025.

[5] Ben Goertzel. *Practical Intelligence Lives on the Pareto Frontier Between Independence Enhancement and Dependency Modeling*. Draft manuscript.